

Quantum-gates perspective: $\text{SO}(10)$ gauge anomalies from the quantum defect sector

Jusang Lee*

Independent researcher, Republic of Korea

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GAP-3 ($\text{SO}(10)$ anomaly cancellation) is examined from the quantum-gates perspective: under the $\text{SO}(10)$ embedding (Math157) derived from the BCC condensate, the spinor representation **16** contains the precise SM fermion content required for *symbolically exact* cancellation of the five SM perturbative gauge anomalies and the mixed gravitational $\text{Tr}(\text{U}(1)_Y)$ anomaly per generation. The right-handed neutrino singlet **1**(5) has SM hypercharge $Y_{\text{RHN}} = 0$ (Eq. (4) below) and therefore contributes *zero* to any $\text{U}(1)_Y$ trace; the cancellation is intrinsic to the $\bar{\mathbf{5}} \oplus \mathbf{10}$ SM matter content alone. Numerical evaluation cross-checks the symbolic result at machine precision and is reported as a redundant cross-check, NOT as a finite-precision estimate. The current canonical tier of GAP-3 is **T6 proved conditional** on (H1) the Math157 $\text{SO}(10)$ embedding and on (H2) Pillar 4 hypothesis H1.1 of Math80-AddA; unconditional promotion is *not* performed in this paper.

I. INTRODUCTION

The quantum-completion programme (Math60) specifies four closure gates: GAP-1 (hbar), GAP-2 (BRST), GAP-3 (anomalies), GAP-4 (cosmology). GAP-3 asks: does the quantum defect sector (Pillar 4, gates) support exact quantum anomaly cancellation?

The answer is yes. The $\text{SO}(10)$ spinor representation **16** contains one SM family plus a right-handed neutrino. This precise content is what is required for anomaly cancellation. No fine-tuning; it is geometrically forced.

From the quantum-gates perspective, the anomaly cancellation is a **consequence** of the gauge-group emergence, not an independent constraint. This paper demonstrates that connection explicitly.

* jtkor@outlook.com

II. SETTING AND MAIN RESULT

The $SO(10)$ spinor decomposes under $SU(5) \times U(1)_\chi$ as

$$\mathbf{16} = \mathbf{10}(1) \oplus \bar{\mathbf{5}}(-3) \oplus \mathbf{1}(5). \quad (1)$$

Further breaking to SM:

$$\mathbf{10}(1) \rightarrow (\mathbf{3}, \mathbf{2})_{1/6} \oplus (\bar{\mathbf{3}}, \mathbf{1})_{-2/3}, \quad \bar{\mathbf{5}}(-3) \rightarrow (\mathbf{1}, \mathbf{2})_{-1/2} \oplus (\bar{\mathbf{3}}, \mathbf{1})_{1/3}. \quad (2)$$

Theorem 1 ($SO(10)$ anomaly cancellation: symbolic exactness). *Under hypothesis (H1) the $SO(10)$ embedding theorem (Math157), the $SO(10)$ spinor $\mathbf{16}$ contains the exact SM fermion content required for symbolic anomaly cancellation. The relevant anomaly coefficients to verify (under standard SM perturbative-anomaly classification) are:*

- $\text{Tr}(U(1)_Y^3)$ — pure hypercharge perturbative.
- $\text{Tr}(SU(3)_c^2 \cdot U(1)_Y)$ — mixed colour–hypercharge.
- $\text{Tr}(SU(2)_L^2 \cdot U(1)_Y)$ — mixed electroweak–hypercharge.
- $\text{Tr}(U(1)_Y)$ — mixed gravitational–hypercharge (gravitational anomaly trace).

The $SU(3)_c^3$ anomaly vanishes by vector-like quark content; the $SU(2)_L^3$ perturbative anomaly is group-theoretically trivial, with the relevant constraint being the Witten global $SU(2)$ anomaly (treated separately). All four non-trivial coefficients vanish symbolically (exact algebraic zero); numerical evaluation cross-checks at machine precision. The result is conditional on Math157 $SO(10)$ embedding; Math281 records the algebraic-closure observation that the cancellation, once the $SO(10)$ embedding is fixed, is automatic.

III. PROOF SKETCH

a. Step 1 ($SO(10)$ structure and spinor decomposition) — Math157 derives the complete decomposition of the $SO(10)$ spinor under all intermediate gauge groups. The decomposition is the standard one, familiar from GUT literature.

b. Step 2 (Trace-method computation) — For each anomaly coefficient $\text{Tr}(G^n)$, compute

$$\text{Tr}(G^n) = \sum_f T_f^{(n)} Q_f, \quad (3)$$

where the sum is over all fermions in the spinor, and $T_f^{(n)}$, Q_f are Dynkin indices and charges. Math157 tabulates all 36 Dynkin indices (6 anomaly types $\times$ 6 fermion fields) and verifies that the six sums all equal zero.

c. Step 3 (Right-handed neutrino role: $Y = 0$, NOT $Y = 1/2$). Physics fact (Math314-AddD physics audit, binding). The $\text{SO}(10)$ right-handed neutrino sits in the $\mathbf{1}(5)$ component of the $\mathbf{16} \rightarrow \mathbf{10}(1) \oplus \bar{\mathbf{5}}(-3) \oplus \mathbf{1}(5)$ decomposition under $\text{SU}(5) \times \text{U}(1)_\chi$. The label “(5)” attached to the singlet is the $\text{U}(1)_\chi$ charge in the standard normalisation $Q_\chi(\mathbf{1}) = +5$; it is *not* the SM hypercharge Y , and it is *not* half the SM hypercharge.

The standard SM hypercharge assignment via $Y = T_{3R} + \frac{1}{2}(B - L)$ on the $\text{SO}(10)$ $\mathbf{16} \supset \mathbf{1}(5)$ singlet gives

$$\boxed{Y_{\text{RHN}} = 0} \quad (4)$$

because the right-handed neutrino has $T_{3R} = +\frac{1}{2}$ and $B - L = -1$ (lepton, L -conjugate). Therefore the RHN *does not contribute directly to any SM $\text{U}(1)_Y$ trace $\text{Tr}(Y^n)$* ; it makes *zero* contribution to $\text{Tr}(\text{U}(1)_Y^3)$, $\text{Tr}(\text{U}(1)_Y \cdot \text{SU}(2)_L^2)$, $\text{Tr}(\text{U}(1)_Y \cdot \text{SU}(3)_C^2)$, and to $\text{Tr}(\text{U}(1)_Y)$. The cancellation of all five SM perturbative gauge anomalies and of the mixed gravitational $\text{Tr}(\text{U}(1)_Y)$ anomaly within each generation of the $\mathbf{16}$ is therefore *intrinsic to the $\bar{\mathbf{5}} \oplus \mathbf{10}$ SM matter content alone*; the RHN $\mathbf{1}(5)$ does not participate in any of these cancellations.

The genuine consistency role of the RHN is elsewhere: (i) algebraic completion of the $\text{SO}(10)$ $\mathbf{16}$ multiplet so that the $\text{SO}(10)$ gauge group itself is anomaly-free ($\text{SO}(10)$ is anomaly-free as a simple Lie algebra; the $\mathbf{16}$ is the spinor that realises this); (ii) closure of the gauged $\text{U}(1)_{B-L}$ mixed gravitational anomaly $\text{Tr}(B - L)$ when $B - L$ is gauged in the $\text{SO}(10)$ extension (the RHN carries $(B - L) = +1$ and is required to make $\text{Tr}(B - L) = 0$ per generation); (iii) Witten $\text{SU}(2)_L$ global anomaly bookkeeping (the RHN is an $\text{SU}(2)$ singlet, contributes zero doublets, and is therefore neutral with respect to the Witten constraint). The Math314-AddD audit identified an earlier draft mis-statement $Y_{\text{RHN}} = 1/2$, which is incorrect; the corrected (4) is the binding canonical value.

d. Step 4 (Conditional algebraic closure under (H1)+(H2)). Math281 records the algebraic-closure observation that, conditional on (H1) Math157 $\text{SO}(10)$ embedding and (H2) Pillar 4 H1.1 (Math80-AddA), the anomaly cancellation is an algebraic identity: the anomaly coefficients either vanish symbolically (as they do for the SM $\bar{\mathbf{5}} \oplus \mathbf{10}$ content) or do not vanish at all; there is no

continuous family of solutions. The verdict therefore inherits the Pillar 4 conditionality and is recorded at **T6 proved conditional** on (H1)+(H2). Promotion to T7 PROVED (unconditional) requires the discharge of (H1) and (H2) independently and is *not* performed in this paper.

IV. SYMBOLIC VERIFICATION (NUMERICALLY CROSS-CHECKED)

The relevant anomaly coefficients computed via the spinor-trace method (Math157) are *symbolically exact zero*; numerical evaluation cross-checks at machine precision (Math319 V1 acceptance gate). The coefficient table is:

$$\text{Tr}(\text{U}(1)_Y^3) \equiv 0 \quad (\text{symbolic exact}) \quad (5)$$

$$\text{Tr}(\text{SU}(3)_c^2 \cdot \text{U}(1)_Y) \equiv 0 \quad (\text{symbolic exact}) \quad (6)$$

$$\text{Tr}(\text{SU}(2)_L^2 \cdot \text{U}(1)_Y) \equiv 0 \quad (\text{symbolic exact}) \quad (7)$$

$$\text{Tr}(\text{U}(1)_Y) \equiv 0 \quad (\text{symbolic exact, gravitational anomaly trace}) \quad (8)$$

$$\text{Tr}(\text{SU}(3)_c^3) \equiv 0 \quad (\text{vector-like quark content}) \quad (9)$$

$$\text{Tr}(\text{SU}(2)_L^3) \equiv 0 \quad (\text{group-theoretically trivial; Witten anomaly}) \quad (10)$$

treated separately as a global anomaly). (11)

The cancellations are exact at the symbolic level; the numerical “six-digit precision” wording of an earlier draft has been corrected per CLAUDE.md §15.6 rule #7 (theorem-vs-estimate notation discipline) and aligned with Math319 V1 acceptance gate of symbolic exact zero with numerical cross-check.

V. DISCUSSION — CONDITIONAL GAP-3 VERDICT, CURRENT CANONICAL STATUS OF THE FOUR GAPS

From the quantum-gates perspective, Pillar 4 (gauge emergence) fixes the $\text{SO}(10)$ embedding structure. Conditional on (H1) the Math157 $\text{SO}(10)$ embedding theorem (and on Pillar 4 H1.1 of Math80-AddA), the SM perturbative gauge anomalies and the mixed gravitational $\text{Tr}(\text{U}(1)_Y)$ anomaly cancel *symbolically exactly* per generation, by intrinsic $\bar{\mathbf{5}} \oplus \mathbf{10}$ content of the $\text{SO}(10)$ **16**, without any direct Y -charge contribution from the RHN singlet (Eq. (4)).

a. Current canonical status of the four GAP gates. Per Math307 / Math310 (canonical, post-Math314 audits), the present canonical tiers are

$$\text{GAP-1} = T4, \quad \text{GAP-2} = T6, \quad \text{GAP-3} = T6, \quad \text{GAP-4} = T3.$$

The Stage-2 composite is therefore $\min = T3$ (driven by the GAP-1 verdict-shell consumption pending in Math312 and the GAP-4 falsifiability-package readiness in Math311). The earlier draft wording “GAP-1, GAP-2, and GAP-3 all pass inspection” is withdrawn; the correct status is *GAP-2 and GAP-3 conditional on Pillar 4 H1.1 + companion hypotheses; GAP-1 at $T4$ with verdict-shell pending; GAP-4 at $T3$.*

b. What the present paper records. A symbolically exact, numerically cross-checked verification of the five SM perturbative gauge anomalies and the $\text{Tr}(\text{U}(1)_Y)$ gravitational mixed anomaly per generation, conditional on (H1), with the explicit RHN hypercharge (4) fixing the misstatement of an earlier draft. The paper does *not* certify Stage-2 closure and does *not* promote any other GAP tier; it is the GAP-3 audit- auxiliary note recorded against the Math319 V1 verification target.

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